



Building an AI Strategy

From Questions to Action: NJDOT's Journey

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— BUREAU OF —
RESEARCH AND INNOVATION



AI Background

What is Artificial Intelligence (AI)?

AI learns from data to make decisions and perform tasks that normally need human intelligence

Types of AI

- Machine Learning
- Deep Learning
- Natural Language Processing
- Generative AI
- Agentic AI
- Natural Language Processing (NLP)
- Computer Vision
- And other

How AI Works

1. Identify Problem
2. Collect Data
3. Prepare Data for AI
4. Train the System
5. Test & Improve
6. Deploy According to Policies and Standards, Learn and Maintenance

What we have now access at NJDOT

- ☐ Copilot
- ☐ NJ AI Assistant
- ☐ AI Development Plat Forms



Where We Started

Our Challenge in 2021

The Big Question: "Does AI actually work for DOT functions—beyond drafting emails and summarizing documents?"

What We Didn't Know:

- Which AI capabilities were real vs. hype
- Who should own AI initiatives
- Whether our data was even usable for AI
- How to get leadership buy-in
- Where to start without a big budget



NJDOT's AI Journey - Key Milestones

From Exploration to Implementation:

2021-2022: Research & Discovery

- Monitored federal/state AI policies
- Studied AI applicability in transportation
- Built foundational knowledge



2023: Foundation Building

- First NJ State AI Policy released
- NJ AI Assistant tool launched
- Started internal education



2024: Organization & Collaboration

- Established NJDOT AI Working Group
- Joined national peer exchanges
- Began use case identification



2025: Action & Pilots

- Selected two pilot working groups (Construction/Materials, Bridge)
- Conducting research pilots
- Developing prioritization framework



2026: Strategic Expansion

- Leading TPF-5(573) AI Pooled Fund Study
- Developing NJDOT AI Strategic Plan (IT is lead)



Action Framework

PHASE 1: Educate & Engage

- ☐ Build AI literacy, identify champions
- ☐ Educate workforce + Use case list

PHASE 2: Pilot & Learn

- ☐ Small-scale pilots, measure outcomes
- ☐ ROI + Lessons learned

NJDOT AI Plan

PHASE 3: Assess & Prioritize

- ☐ Audit data readiness for full implementation
- ☐ Prioritize successful pilot projects + Data gaps map

PHASE 4: Scale & Formalize

- ☐ Strategic plan, enterprise deployment
- ☐ AI governance + Scaled implementations



Build AI Literacy

We Can't Build AI Strategy Without Understanding

What We are doing:

- **Lunch & Learn Series**
- **Cross-Departmental Workshops/** Unit-specific brainstorming sessions
- **Executive Briefings** - on AI capabilities/risks
- **Working Group Formation** - Identified AI champions across units

Unexpected Discovery:

Staff were MORE excited about AI possibilities than we expected



Build AI Literacy

Activities:

- Started with "AI types relevant to DOTs" (Computer Vision, NLP, Predictive Analytics)
- Showed real examples from other state DOTs
- Encouraged "what if" thinking in each unit
- Made it low-pressure exploration, not a mandate

Investment Required: Minimal—mostly staff time (8-10 hours/month)

Outcome: 25+ units identified potential use cases; created foundation for buy-in



Designing Pilots - Structure

Pilot Components:

CLEAR PROBLEM DEFINITION

- Specific task AI will perform
- Current process baseline metrics
- Success criteria defined upfront

APPROACH

- Phase 1: Data prep & model training
- Phase 2: Testing & validation
- Phase 3: Limited production use
- Phase 4: Evaluation & documentation

DEDICATED TEAM

- Subject Matter Expert (owns the work)
- AI Liaison (AI expertise)
- IT Partner (infrastructure)

DOCUMENTATION PLAN

- Status updates
- Lessons learned log
- Final report with ROI analysis



Prioritization of use cases for Pilots

Bridge Use Case Ideas	Data Availability for Pilot (Yes/No)	Risk (Low/Medium/High)	Priority Ranking (1/2/3)
1. Factsheets based on Historical Excel file: Training data; Conditions by Ownership; Performance by Deck Area and Count; GIS based queries			
2. Promotional and New Hire Videos/materials: Quick 1 to 2 Minutes narrative			
3. Deep Dive modeling using Bridge Data: General Condition Rating versus Element Condition Rating; Interaction - BMS with other Management Systems such as PMS, PMRS, etc.; Prediction Modeling - Deterioration, Future Projects			
4. Bridge Inspection: Early detection, Image Recognition; Report writing, Recommendation			
5. Bridge Design and Rating			
6. Bridge Design cost predictions for alternatives based on data collected from current projects			
7. Image based defect analysis and repair recommendations			
8. Superstructure and substructure recommendations based on inputs (ie. span length, geometrics, vertical			



Securing Buy-in & Funding

Leadership Buy-in Strategy:

EARLY STAGE:

- Starting with education, not funding requests
- Position AI as force-multiplier, not replacement
- Showing examples from peer DOTs (relatable proof)
- Emphasize risk management (what if we don't explore this?)

PILOT STAGE:

- Identify quick wins that needed minimal investment
- Partner with existing IT infrastructure/tools (Copilot, NJ AI Assistant and other)
- Use Research budget for exploratory pilots
- Engage champions at multiple organizational levels

SCALING STAGE:

- Demonstrate ROI from pilots
- Develop business cases tied to strategic goals
- Propose AI strategic plan with phased investment
- Leverage AI pooled fund study



TPF- 5(573): State DOT Council for Strategic AI Adoption

Pooled Fund Study Objectives:

- Biannual peer exchanges to share AI experiences and explore opportunities
- Develop actionable AI deployment strategies
- Collaboration with FHWA on national level AI strategy
- Multi-state pilot projects on common use cases
- Standardize data quality to enable AI-ready datasets
- Create shared resources: research repositories, toolkits, training materials

TPF TRANSPORTATION POOLED FUND

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Transportation Pooled Fund - Study Detail

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State DOT Council for Strategic AI Adoption

Print

General Information	Financial Summary
Study Number: TPF-5(573)	Contract Amount:
Former Study Number:	Suggested Yearly Contribution: \$20,000.00
Lead Organization: New Jersey Department of Transportation	Total Commitments Received: \$1,445,000.00
Solicitation Number: 1653	100% SP&R Approval: Approved
Partners: CA, CT, ID, KY, MA, MDOT SHA, MN, NC, NJ, OK, PADOT, TX, UT, VA, VT, WA, WI, WV	
Status: Cleared by FHWA	
Est. Completion Date:	
Contract/Other Number:	
Last Updated: Mar 30, 2026	
Contract End Date:	

Contact Information
Lead Study Contact(s): Giri Venkiteela Giri.Venkiteela@dot.NJ.gov Phone: 6099632239
FHWA Technical Liaison(s): Craig Thor Craig.Thor@dot.gov Phone: 202- 493-3338

- Commitments by Organizations
- Study Description
- Documents
- Solicitation Documents



TPF-5(573): State DOT Council for Strategic AI Adoption

What are States getting:

- Learn from other State's pilots (successes AND failures)
- Access to shared implementation toolkits
- Workforce training materials
- Latest information on AI technology
- No need to reinvent the wheel on common use cases

Current Status:

- Steering Committee established
- Kickoff meeting: 2nd September 2026
- Planning on first Peer exchange



Thank you.

Questions?